

Case Study: Design of Test Cases and a Maintenance Plan(Bus Reservation System)

Application: Bus Reservation System

Introduction

This case study presents test case design and a maintenance plan for a Bus Reservation System.

Test Cases

Test ID	Module	Test Scenario	Expected Result
TC01	Login	Valid username/password	Login successful
TC02	Login	Invalid password	Error message
TC03	Search	Search available buses	Matching buses displayed
TC04	Booking	Book available seat	Booking confirmed
TC05	Payment	Successful online payment	Ticket generated
TC06	Cancellation	Cancel booked ticket	Cancellation and refund initiated
TC07	Admin	Add new bus	Bus added successfully
TC08	Reports	Generate booking report	Report displayed

Testing Types

- Unit Testing
- Integration Testing
- System Testing
- User Acceptance Testing (UAT)
- Performance Testing

- Security Testing

Maintenance Plan

Corrective Maintenance

- Fix software bugs, payment issues, and login errors.

Adaptive Maintenance

- Update software for new operating systems, browsers, and payment gateways.

Perfective Maintenance

- Improve user interface, optimize search speed, and add new features.

Preventive Maintenance

- Database backup, security patches, code refactoring, and server monitoring.

Maintenance Schedule

Activity	Frequency
Bug Fixes	Weekly
Security Updates	Monthly
Database Backup	Daily
Performance Review	Monthly
Feature Updates	Quarterly

Conclusion

Well-designed test cases ensure reliable functionality, while a structured maintenance plan improves security, performance, and long-term usability of the Bus Reservation System.